SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA0408**

COURSE OUTCOME COVERED

CO2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage: 15%

Assessment Tool Weightage: 20%

QUESTIONS: **Total Marks: 20**

1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.

2. Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.

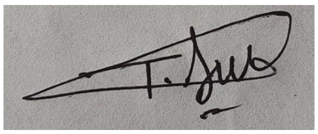
3. Compare ReLU, Leaky ReLU (LReLU) and ELU (Exponential Linear Unit) activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.

4. Explain the concept of unsupervised learning in neural networks. Describe the working principle of Restricted Boltzmann Machines (RBMs) and Autoencoders and compare their objectives and applications.

Code of Conduct:

I T. Narasimha(192425233) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developin g**	**Level 1 – Beginning**
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

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	Aspect	Machine Learning	Deep Learning
1	Architecture	Usess traditionae argorithms suchas Decision Trees supper vector machines Random forest, 4. Linear Regression	Ues ANN with multiple hidden players
2.		Can produce good results with small tomedium-sized datasets.	Requires very large amounts of data to achieve high.
3.		Feature manually selected and enginer by experts before traning the model	Automatically learns and extract;
4.		Suitable for structured data suchas Costomer records financial data & sales data.	Best suited for unstructured data such as images.

Real world Applications of Deep Learning

1. Face recognition systems used in smartphones, security systems and attendance systems to identify individuals accurately.

2. Autonomous cars - Deep-Learning enables vehicles to recognise roads, traffic signs, ~~(presed)~~ Pedestrains And other vehicles for safe navigation.

⇒ Real world Applications of Machine Learning

1. Email Spam Detection - ML algorithms classify incomming emails as spam (or) legitimate based on patterns in the mail Contact.

2. Credit card Fraud Detection Banks use ML models to identify suspicious transactions and Prevent fraudlent activities.

CONCLUSION

Machine Learning is effective for solving problems with structured data and limited datasets while deep learning excels at handling complex tasks involving images, speech and text by automatically learning features from large volumes of data.

2. Representation Learning is a technique in Deep Learning where the model automatically learns useful features from raw data instead of relying on manually designed features. It identifies Patterns and creates meaningful representation of the input data that help to improve prediction accuracy.

The width of a neural network refers to the number of neurons in each hidden layer while the depth refers to the number of hidden layers in the network. Both play a important role in determining the networks performance

A wide neural network has more neurons in each layer, enabling in to learn a larger variety of features at the same level. This can improve accuracy for simpler tasks but also increases the number of parameters leading to higher memory usage & computational cost.

A deeper neural networks contains more hidden layers, allowing it to learn features in hierarichal manner. More The first layer learn simple features such as edges (or) basic patterns while deeper layers combine them into more complex tasks like image recognition and natural language processing. However they require more training time, greater computational resources and larger datasets. very deep networks may also face issues.

In terms of learning capability, deeper networks are usually more powerful because they can model highly complex relationship in data, whereas wider networks mainly improve the model's capacity to learn more features with a single layer. The width of a network increases its ability to learn more features within a layer while the depth allows it to learn complex & abstract representation.

3.a. ReLU

$f(x) = \max(0, x)$

Returns 0 for negative inputs and x for positive inputs

Leaky ReLU

$$f(x) = \begin{cases} x, & x > 0 \\ \alpha x, & x \leq 0 \end{cases}$$

Here α is a small constant allowing a small negative output instead of zero

ELU

$$f(x) = \begin{cases} x, & x > 0 \\ \alpha(e^{x}-1) & x \leq 0 \end{cases}$$

Produces smooth negative values using an exponential function, which helps maintain non-zero gradient

ReLU (Advantages)

- Easy to implement and computationally efficient
- Speedup model training
- Reduces the vanishing gradient problem

LReLU

- solves the dying ReLU problem
- maintains gradient flow even for negative inputs.

ELU

- provides smooth and continuous outputs
- Faster convergence during training.
- Improves model stability by producing outputs

CONCLUSION

ReLU is the most widely used activation function because of its simplicity and efficiency. Leaky ReLU improves upon ReLU by preventing neurons from becoming inactive, making it more suitable deeper networks.

4 # Introduction

Artificial intelligenc enables mathines to Perform task such as learning reasoning prediction and decision making. One of the important branches. Amog these unsupervised learning is widely used when labeled data is unavailable instead of learning from predefined target outputs.

Unsupervised learning in Neural Network.

Unsupervised learning is a type of machine learning in which the neural network is trained using unlabeled data.

characteristic of unsupervised learning

* Uses unlabeled dataset
* Automatically discover hidden structure
* Helps in anomaly detection
* Useful when labeled data is unavailable

working process of unsupervised learning

Raw unlabeled data

⇓

Data preprocessing

⇓

Neural Network

⇓

Pattern Discovery

⇓

Feature learning

⇓

Useful Representation.

Advantages

* Does not require labeled dat

* Reduce manual effort

* Discover hidden relationship

* Useful for large data set

* perform automatic feature learning